# Driving Behavior in Work Zone

## Informed Safety, Mobility, and Driver Comfort Enhancement Practices for Work Zones: Learnings from High-Fidelity Data

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### Abstract

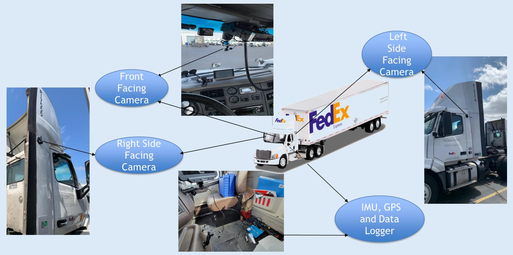
This study analyzes the impacts of work zones (WZ) on safety, mobility, and comfort along I-40 using high-fidelity data. Data from WZs near Jackson, Tennessee (MM 79, 93.4) show speed drops, variability, and directional shifts, highlighting risks and discomfort.

### Objectives

- Review factors affecting safety, mobility, and comfort at WZs.
- Define performance measures for quantifying safety, mobility, and comfort.
- Collect high-fidelity vehicle dynamics data using sensors.
- Calculate measures from speed, acceleration, heading, and spacing data.

### Data Collection

- Used GPS (10 Hz) and IMU (10 Hz) on a test vehicle for a pilot test.
- Captured position, speed, acceleration, and heading via sensors using FedEx trucks in I-40.
- Captured surrounding vehicles using video cameras on front, and two sides.

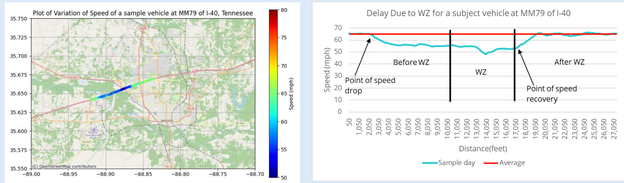



Instrumental Set-Up for Data Collection.

Samples of the image frames processed using YOLOv8 and ByteTrack.

### Data Processing and Analysis

- Synced GPS (10 Hz) and IMU (10 Hz) using timestamps.
- Segmented stretch of WZ area: From 2 miles before to 2 miles after WZ.
- Sensor data and video data clipped and processed to study the behavior in WZ
- Combination of data used to obtain the position of surrounding vehicle.




Speed variation (left) and Delay (right) due to the Work Zone.

### Findings

- Vehicles reduce speed upon approaching WZs. (Nearly 10 mph)
- Speed variability increases, indicating driver discomfort.
- Speed drop is more significant in WZs, indicating delays.
- Frequent lane changes and abrupt braking are prevalent in WZs.
- Fusion of video data and IMU data provides information on surrounding vehicles.

### Future Steps

- Analyze the captured data in non-WZ areas as well.
- Study of near-crash events for the trucks.
- Investigate potential cost savings (crash reduction, less downtime), safety improvement (protecting drivers and cargo), and efficiency gains (faster deliveries).

Acknowledgement: We would like to acknowledge FedEx for facilitating the data collection process by providing us access to the stations and installation of the sensors on delivery trucks.

## Strategies for Improved Driver Behavior within Work Zones

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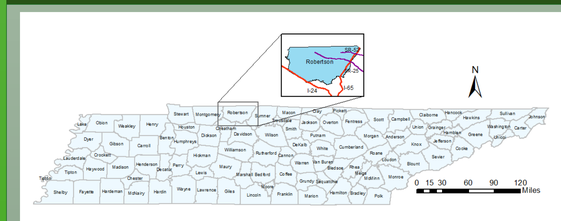
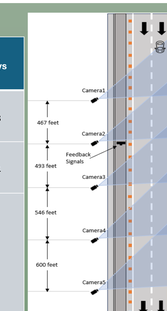
### Abstract

This study analyzes the driving behavior in a long WZ in Robertson County on I-65. Southbound traffic was captured. Five cameras were installed for data collection, each about 500 feet apart. Continuous data collection was made for nearly 2 months. Dynamic Speed Feedback System (DSFS), Speed Wizard (SW) and Combination (SW and PCMS) was employed to study the driving behavior.

### Study Area and Duration




| Equipment | Date | Action (Installed/ Removed) | Message Shown | Days |
|---|---|---|---|---|
| DSFS | 10/3/2023 | Installed | "YOUR SPEED XX" | 13 |
| | 10/16/2023 | Removed | | |
| SW | 10/26/2023 | Installed | "SPEED LIMIT 55, YOUR SPEED XX" | 14 |
| | 11/9/2023 | Removed | | |
| SW and PCMS | 11/13/2023 | Installed | "SPEED LIMIT 55, YOUR SPEED XX" (SW) "SLOW DOWN WORKER AHEAD" (PCMS) | 8 |
| | 11/21/2023 | Removed | | |

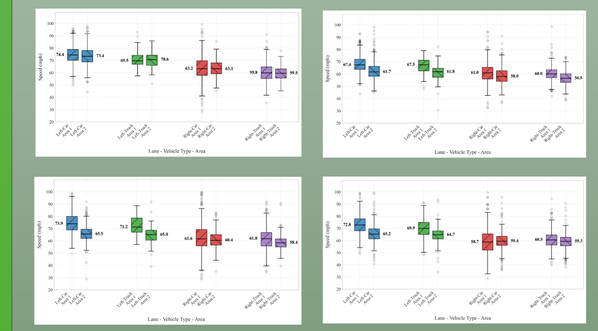



### Analysis 1: Generalized Ordered Logit

The data collected by the last camera over nearly 14 different days, each one hour, was used for analysis. The speed data was divided into three categories based on violations above the speed limit. The Generalized Ordered Logit model was used to model the different speed violations. Some of the findings are:

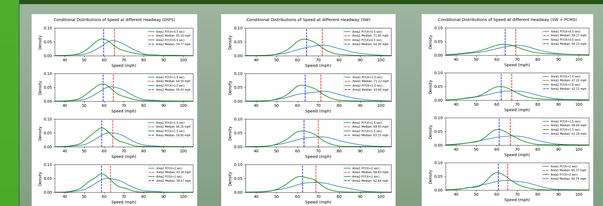
- Speed violations are more common on cars, in left lanes, on weekends, and in non-inclement weather conditions.
- Speed feedback mechanisms help reduce speeding violations and increase the percentage of speed limit compliance.
- If a vehicle is following another vehicle too closely (time), then it might be violating the speed limit.

### Observation: Speed Distribution



Speed Distribution for camera 1 and camera 5, for Base (Top Left), DSFS (Top Right), SW (Bottom Left), Combined (Bottom Right)

### Observation: Joint Distribution



Speed Distribution for camera 1 and camera 5, for DSFS (Left), SW (Middle), Combined (Right)

### Analysis 2: Generalized Additive Model

The data collected by the first camera and the last camera over different days were used to model a Bayesian Generalized Additive Model to study the relationship between the effect of the speed feedback mechanisms and the speed drop between the two areas. Some of the findings are as follows:

- Speed before feedback is nonlinearly affected due to feedback. It may increase or decrease based on whether the speed is above 61 mph or below.
- The presence of extra PCMS with SW does not yield a significant result as compared to DSFS or SW alone.
- SW decreased speed by 3 mph on average.
- Being on the left lane, a car, and the weekend are all associated with high traffic speed.
- Based on the findings:
  - Policy should focus cars on the left lane on weekends.
  - The frequency of SW or DSFS at some interval decreases speed limit violations and speed variance.